knife-edge, you will remember, in the story of the grudgers and cheats in Chapter 10.

It obviously matters, therefore, on which side of the knife-edge a population happens to *start*. And we need to know how it might happen that a population could occasionally cross from one side of the knife-edge to the other. Suppose we start with a population already sitting on the Always Defect side. The few Tit for Tat individuals don't meet each other often enough to be of mutual benefit. So natural selection pushes the population even further towards the Always Defect extreme. If only the population could just manage, by random drift, to get itself over the knife-edge, it could coast down the slope to the Tit for Tat side, and everyone would do much better at the banker's (or 'nature's') expense. But of course populations have no group will, no group intention or purpose. They cannot strive to leap the knife-edge. They will cross it only if the undirected forces of nature happen to lead them across.

How could this happen? One way to express the answer is that it might happen by 'chance'. But 'chance' is just a word expressing ignorance. It means 'determined by some as yet unknown, or unspecified, means'. We can do a little better than 'chance'. We can try to think of practical ways in which a minority of Tit for Tat individuals might happen to increase to the critical mass. This amounts to a quest for possible ways in which Tit for Tat individuals might happen to cluster together in sufficient numbers that they can all benefit at the banker's expense.

This line of thought seems to be promising, but it is rather vague. How exactly might mutually resembling individuals find themselves clustered together, in local aggregations? In nature, the obvious way is through genetic relatedness—kinship. Animals of most species are likely to find themselves living close to their sisters, brothers, and cousins, rather than to random members of the population. This is not necessarily through choice. It follows automatically from 'viscosity' in the population. Viscosity means any tendency for individuals to continue living close to the place where they were born. For instance, through most of history, and in most parts of the world (though not, as it happens, in our modern world), individual humans have seldom strayed more than a few miles from their birthplace. As a result, local clusters of genetic relatives tend to build up. I remember visiting a remote island off the west coast of Ireland, and being struck by the fact that almost everyone